

Peer review with AI



Licia Verde

*ICREA & Instituto de Ciencias del Cosmos University of Barcelona
JCAP scientific director, arXiv science advisory council*



Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA



don't

As of today

AI and scientific publishing



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This will be a very different talk from what you are used to

Next talk focuses on producing the contents of the papers

What Understanding Means in AI-Laden Astronomy

Yuan-Sen Ting^{1,2,3*}, André Curtis-Trudel^{4,5}, Siyu Yao⁶

¹Department of Astronomy, The Ohio State University, Columbus, OH, USA.

²Center for Cosmology and AstroParticle Physics (CCAPP), The Ohio State University, Columbus, OH, USA.

³Max-Planck-Institut für Astronomie, Heidelberg, Germany.

⁴Department of Philosophy, University of Cincinnati, Cincinnati, OH, USA.

⁵UC Center for Humanities and Technology, University of Cincinnati, Cincinnati, OH, USA.

⁶Department of Philosophy, School of Humanities, Shanghai Jiao Tong University, Shanghai, China.

*e-mail: ting.74@osu.edu; andre.curtis-trudel@uc.edu; yaosy@sjtu.edu.cn

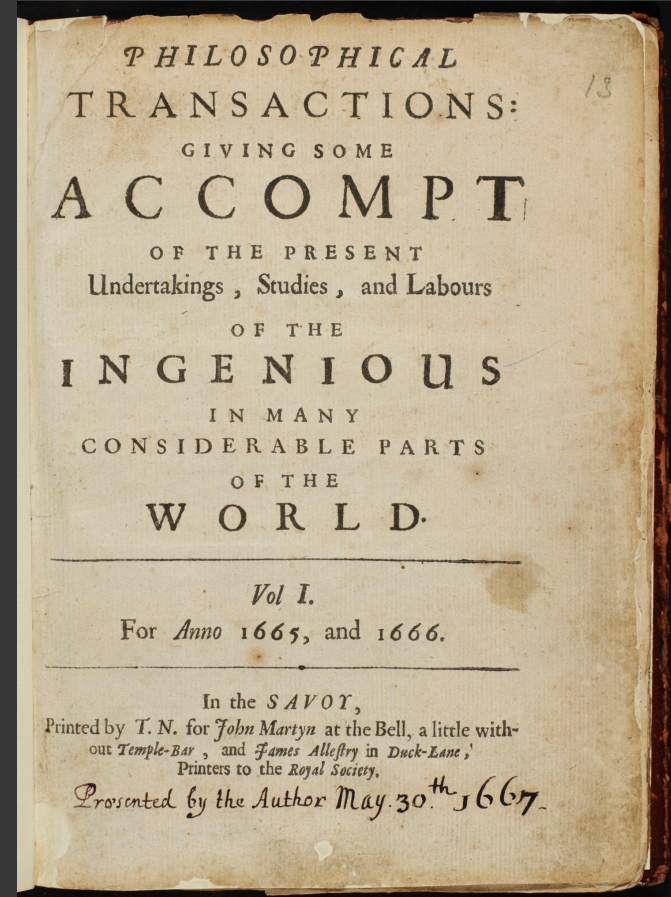
This talk focusses on the scientific publishing aspects

WHAT IS A SCIENTIFIC PAPER?

The modern concept of a scientific paper emerged with the founding of the **Royal Society of London** and its journal ***Philosophical Transactions*** in **1665**



Your
editor...



Before that scientific discoveries were published in “pamphlets” and books (Galileo...)

Then and now

Scientist
(author)



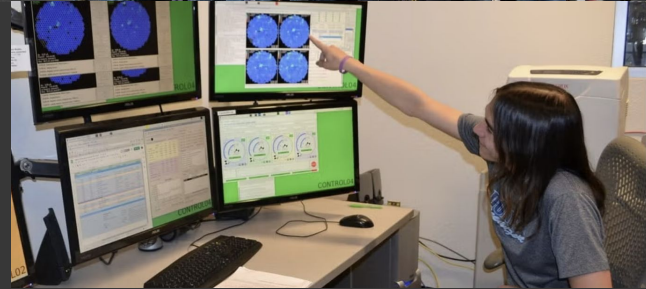
reader



funder



elite



Scale, speed, and complexity; Institutionalization and professionalization; Heavy reliance on technology and collaboration

Today's scientific publishing (Physics, Computer science, Mathematics etc)

Authors do research
and produce paper

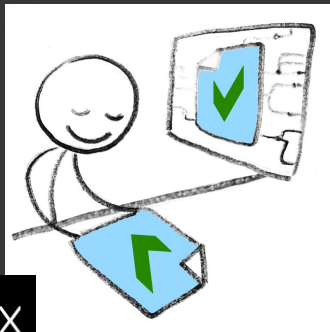


t~months/years



t~hrs/days

Journals
Editors/referees
QA, Curation, refs,
metadata etc.



t~months

Scholarly record
knowledge



Researchers discover new ideas



Results are
"public"



SCIENTIFIC PUBLISHING IS A BUSINESS

BASED ON 'PUBLISH OR PERISH' CULTURE AND 'BIBLIOMETRICS' CREDO

Commercial publishers reach profit margins of 30 to 40% per year
which is higher than that of companies such as Microsoft, Google and Coca Cola

The academic publishing industry has a **financial volume between the music industry and the film industry** (Buranyi S. for the Guardian 2017)

In 2020 annual revenues from article processing charges among major publishers have exceeded 2 billion USD (Zhang et al, 2022, scientometric).
This is three times the UNESCO budget.

All funded by... your grants, overheads, libraries

But arXiv
is free!

Features of academic publishing (vs preprints/arXiv)

✓	Registration	→	Providing a timestamp to officially define who submitted a scientific result first
✓	Dissemination	→	Providing distribution and access to the scientific results, shared publicly
✓	Preservation	→	Preserving the record of science for posterity
✓	Use	→	Promoting and facilitating re-use of scholarly information
✗	Certification	→	Provide a “stamp of approval” with peer review of the submission
✗	Quality Control	→	Provide a high quality product, fully machine readable, w/ cross references, cross listed, findable, accessibility etc.

Filter and
prestige

BIBLIOMETRICS CREDO

What makes a good scientist?

“The metric tide”

Want indicators that are

- Objective
- Quantitative
- Reproducible
- Transparent
- ...

Number of papers

Journal impact factors

Number of citations

GOODHART'S LAW

When a measure
becomes a target
it ceases to be
a good measure

BIBLIOMETRICS CREDO

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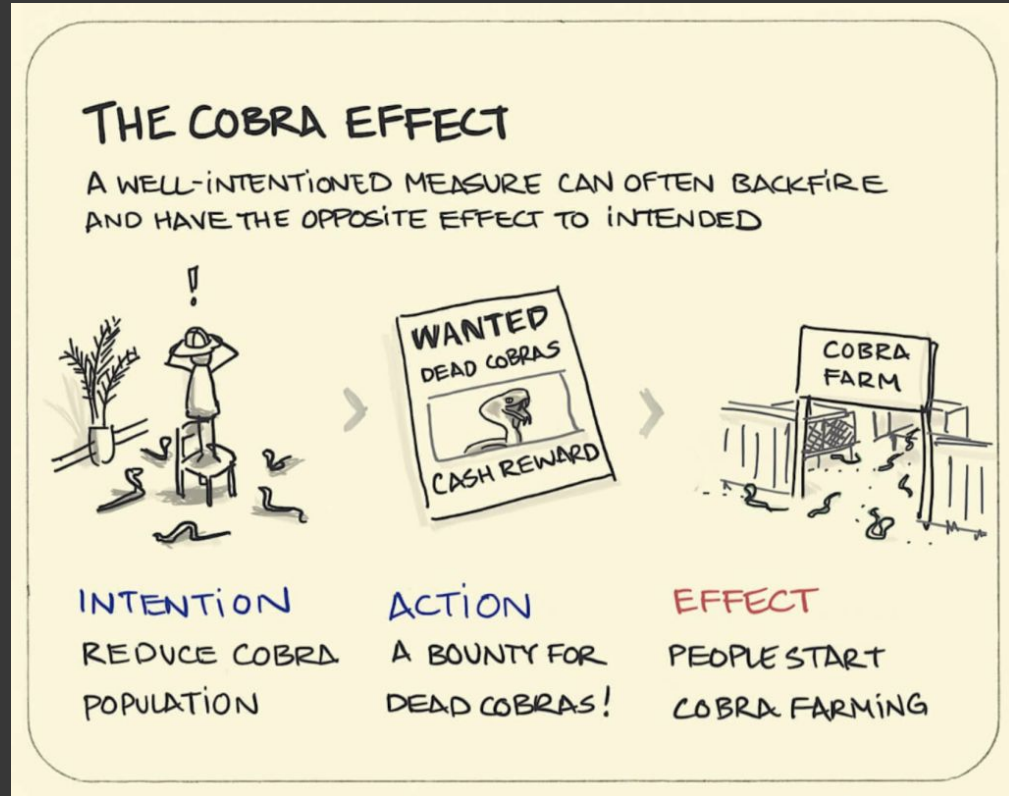
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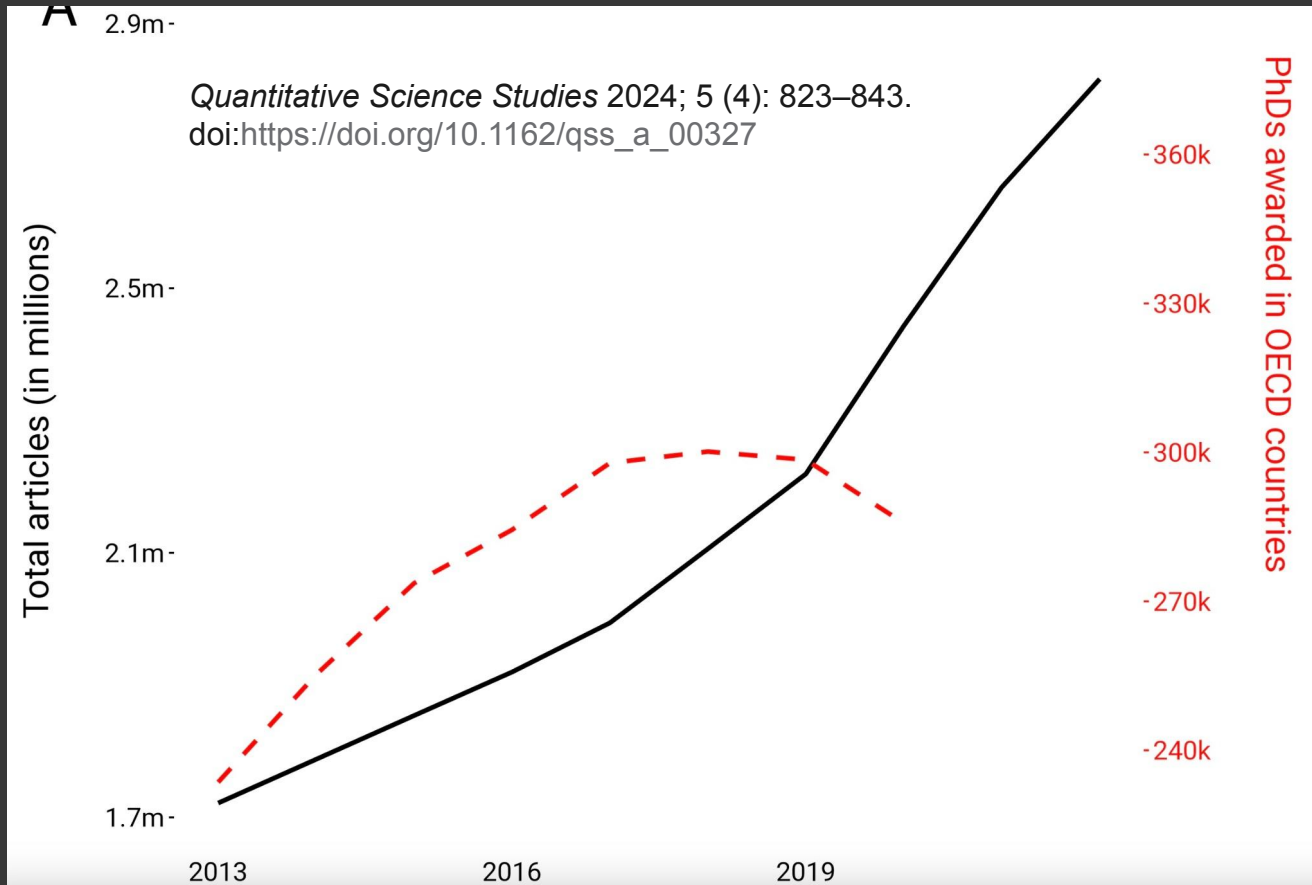
GOODHART'S LAW



THE INCENTIVE SYSTEM (GRANTS, PROMOTIONS, RECOGNITION ETC.) HAS OUTSOURCED THE
RECOGNITION OF QUALITY RESEARCH AND RESEARCH IMPACT TO PUBLISHERS.

USE NUMBER OF PAPERS AND NUMBER OF CITATIONS AS THE CURRENCY OF QUALITY

THE GROWTH OF SCIENTIFIC PUBLICATIONS



Source: N papers -- Scimago website data; N PhDs - OECD

It stays afloat because

Must be $\ll$ of the effort on this

Effort of scientists on this
(non promotable task)

Authors do research
and produce paper

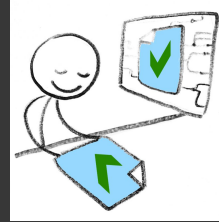


t~months/years



t~hrs/days

Journals
Editors/referees
QA, Curation, refs,
metadata etc.



t~months

Scholarly record
knowledge



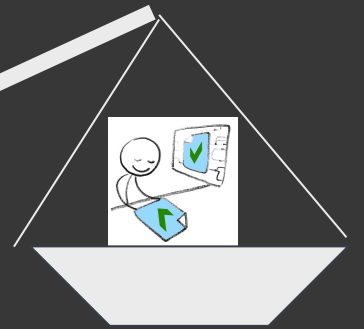
Researchers discover new ideas



Results are
"public"

And content of this
must be of high value:
interest and novelty.
Pristine. Not redundant.
Added value.

Effort balance:

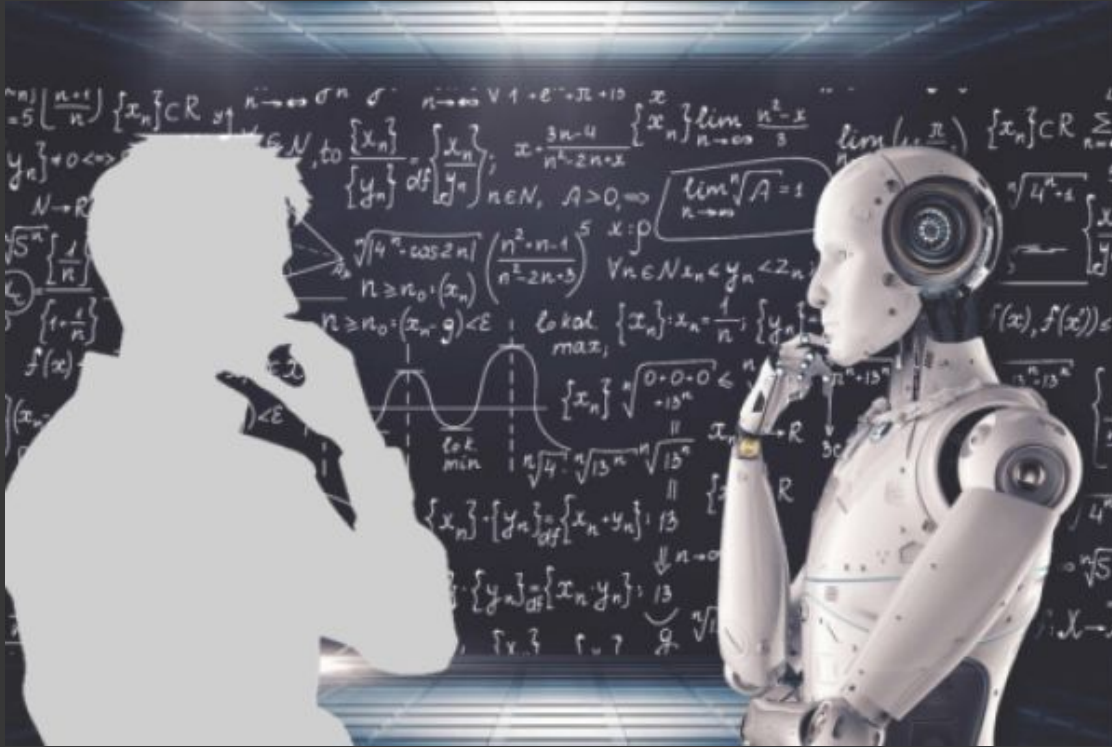


QA, curation etc.

Enables high value of



Like climate change, ignore at your own risk



AI and LLM in scientific publishing can help a lot, improves language presentation, removes barriers,... BUT

AI slop

nonsensical

content looks polished, reads well, but... lacks substance or novelty or ...



THE GROWTH OF DUBIOUS QUALITY SUBMISSIONS

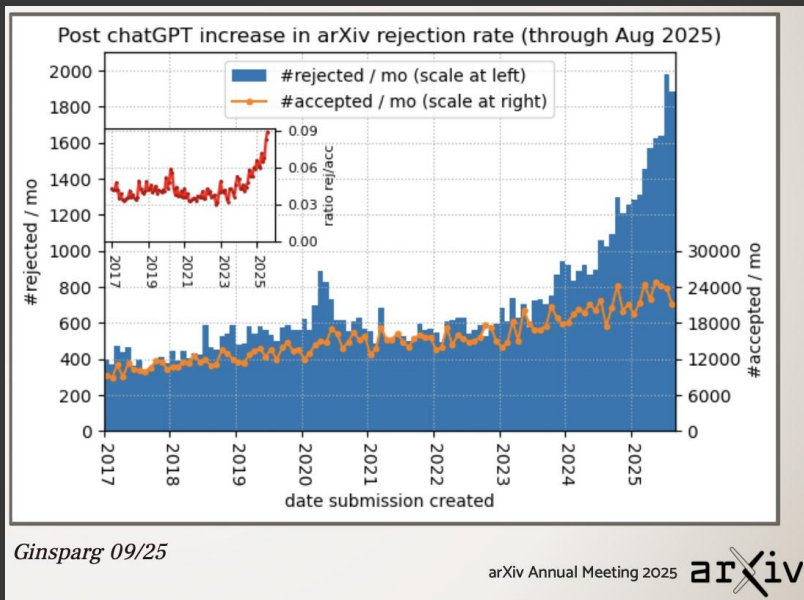
Philosophy

(Louise McNally at Academia Europaea,
Oct '25)

Change 1: S&P's desk rejection rate

Year	Submissions	Desk rejects	%
2020	94	11	11.7%
2021	66	12	18.2%
2022	73	15	20.5%
2023	81	22	27.2%
2024	79	25	31.6%
2025 (ytd)	86	42	48.8%

arXiv aggregated sections



What we detect here is just the tip of the iceberg
AI generated papers have been around and are around across disciplines

2026

SUBMISSIONS GREW BY 40% (COMPARED TO 10-20% AVERAGE YEARS BEFORE)

REJECTIONS DOUBLED (FROM A ~25-30%)

NON SUITABLE 300% INCREASE

symptoms, root causes, or consequences?

REFeree FATIGUE

EDITOR FATIGUE

READERS FATIGUE

SALAMI SLICING

DELAYS

PAPER MILLS

PUBLISH OR PERISH

BIAS TOWARDS POSITIVE RESULTS

PAYWALLS AND ACCESS PROBLEMS

PEER REVIEW, SLOW, INCONSISTENT
SOMETIMES BIASED

JOURNALS IMPACT FACTORS, H-INDEX

METRIC BECOMES A TARGET

FRAGMENTATION AND SPECIALIZATION

INCENTIVES MISALIGNED WITH TRUTH

All there before 2023

SCALE!

LLM like ChatGPT and the like
But now can get very
sophisticated

EXAMPLE THAT BROUGHT IT TO THE OPEN BY DESIGN

Meet Denario, the AI 'research assistant' that is already getting its own papers published

The Denario project: Deep knowledge AI agents for scientific discovery

Boris Bolliet^{*}, Pablo Villanueva-Domingo^{*}, Francisco Villaescusa-Navarro^{*},
Adrian E. Bayer, Aidan Acquah, Chetana Amancharla, Almog Barzilay Siegal, Pablo Bermejo,
Camille Bilodeau, Pablo Cárdenas Ramírez, Miles Cranmer, Urbano L. França, ChangHoon Hahn,
Yan-Fei Jiang, Raul Jimenez, Jun-Young Lee, Antonio Lerario, Osman Mamun, Thomas Meier,
Anupam Anand Ojha, Pavlos Protopapas, Shimanto Roy, David N. Spergel, Pedro Tarancón-Álvarez,
Ujjwal Tiwari, Matteo Viel, Digvijay Wadekar, Chi Wang, Bonny Y. Wang, Licong Xu, Yossi Yovel, Shuwen Yue,
Wenhan Zhou, Qiyao Zhu, Jiajun Zou, Íñigo Zubeldia

^{*}Equal Contribution. Listing order of BB, PVD, FVN is random.



Paper



Code



Docs



App



SCAN ME

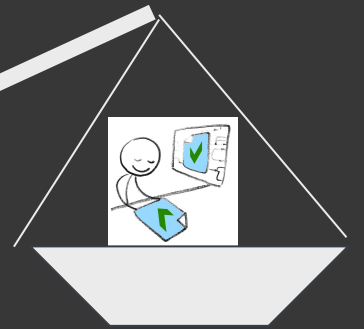
Generates a full paper references included in ~1h

[HTTPS://GITHUB.COM/ASTROPILOT-AI/DENARIOEXAMPLEPAPERS?TAB=README-OV-FILE](https://github.com/ASTROPILOT-AI/DENARIOEXAMPLEPAPERS?tab=readme-ov-file)

It takes you well more than 1h to evaluate and it basically passes all the AI detectors we have available

Anyone can go and use it

Effort balance:

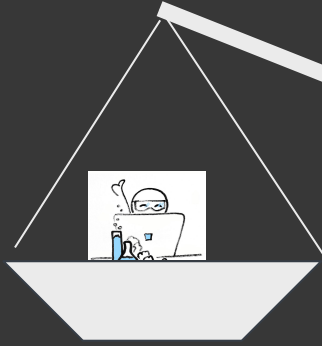


QA, curation etc.

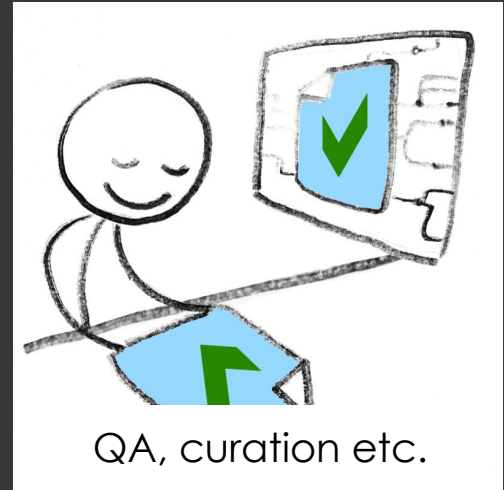
Enables high value of



Effort (un)balance:



Authors produce paper



QA, curation etc.

It can't stay afloat because

Is >> than the effort on this

Effort of scientists on this
(non promotable task)



Authors do research
and produce paper

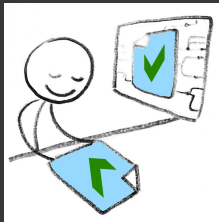


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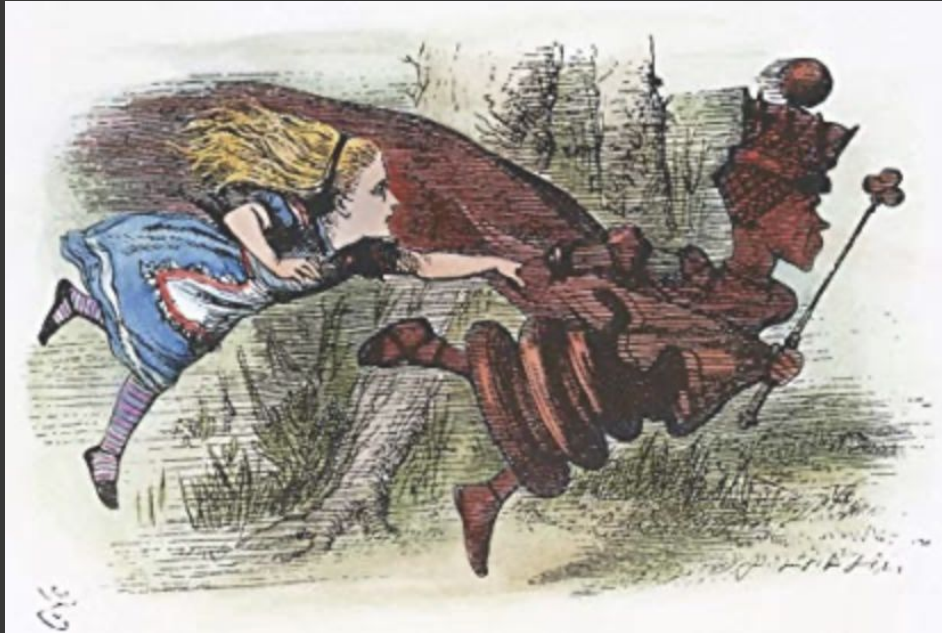
Results are
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Amount of this
must be high value:
interesting. Novelty.
Pristine. Not abundant.
Added value

TOOLS LIKE DENARIO BRINGS IT INTO THE OPEN

The technology is there (whether we like it or not). This is a tool. It will be used.

Current approach→ “Red Queen’s race”

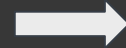


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- **Thomas Kuhn:** Paradigm shifts occur when the old framework can no longer solve the anomalies it faces.(Thomas Meier)



Revolution?

“Technology challenges us to assert our human values, which means that first of all, we have to figure out what they are.” – Sherry Turkle

What is the “atom”, the building block, of (new) scientific knowledge?

What is good research? What is “good researcher”?

What is good research? What is “good researcher”?

NUMBER OF PAPERS

NUMBERS OF CITATIONS



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THE BIG PICTURE

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Eventually we’ll settle in a new system
but it will require a cultural shift
It will not happen overnight

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WHAT CAN WE DO NOW?

Educate editors
Educate referees
Educate authors

Provide AI tools to help with literature search and spot non novel papers
It's possible to make them and really good

Keep the human in the driver's seat

Have open discussions with stakeholders

Ethical Considerations

1. Quality of Research
2. Plagiarism
3. Implications for scientific training and education
4. Assessment of quality and legitimacy
5. Homogenization towards what?
6. Access
7. Authorship
8. Misalignment of incentives in scientific publishing
9. Environmental concerns
10. Malicious use



LLM are not the problem

PEIRIS, NATURE ASTRONOMY 2026

“If the incentive structure of the field rewarded quality rather than quantity, many of the concerns about LLM-assisted paper production would lose their force”.

From my point of view from the trenches of scientific publishing in some sense they are because of the scale. In front of an avalanche of submissions the current system struggles to recognize quality.

The world has changed: e.g.,

(HARVARD) COMPUTERS



1891



today

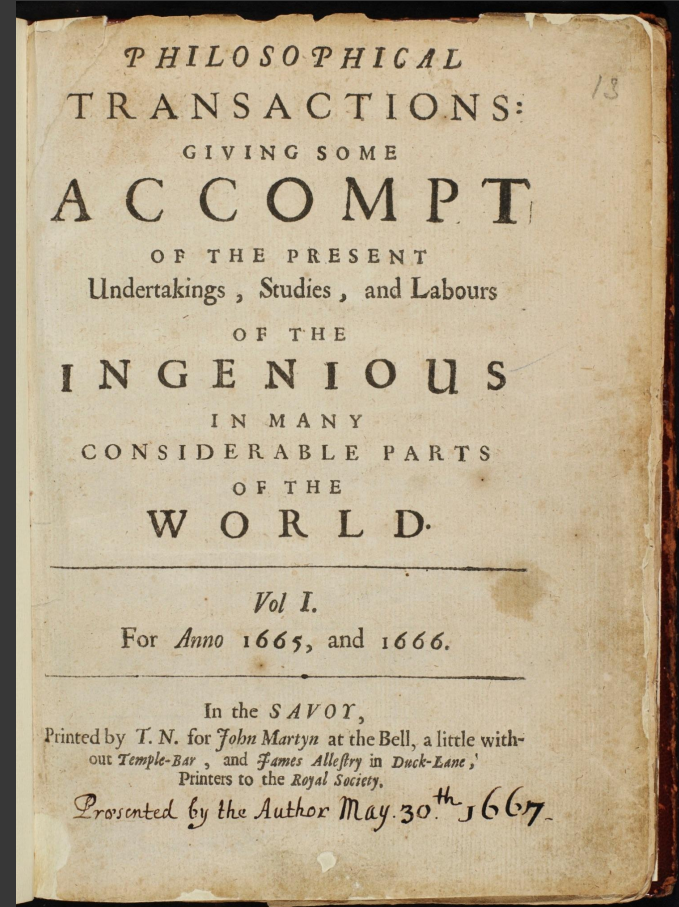
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Your
editor...

The world has changed a lot since then....



In principle:

An LLM can write a great introduction: do we need long introductions –such as today's– be part of "*the scientific paper*"?

AI (focussed on retrieval) can do a very good literature search, provided the databases are machine readable etc. (which increasingly are). Do we need reference lists of hundreds of references as part of "*the scientific paper*"? (or only reference what is directly used)

Papers are becoming extremely long and pedagogical. AI is very good at explaining and at summarizing. Do we need a lot of pedagogical material as part of "*the scientific paper*"?

Radical thought: Anything that a [Claude/chatGPT etc.] can do maybe should not be part of the "*the scientific paper*". The prompts yes. What the author considers a new result yes. Any incremental work that your LMM of choice can do probably not.

Reproducibility becomes a must to increase trust→ open science!

NASA's SciX/ADS have been developing extremely useful tools to curate, find, make accessible publications.

Does “the scientific paper” need to be limited to a non-interactive pdf document?

Is that really the building block/ the atom of scientific knowledge?



Do and don't

FOR AUTHORS

AI cannot be an author because it cannot take responsibility

If you use it (as a tool), declare it

Reproducibility is one of the keystone of science

Authors must take responsibility (ethical issues, plagiarism, attribution, correctness, hallucinations)

FOR REFEREES/EDITORS

AI cannot be a referee because it cannot take responsibility

DO NOT upload the author's document to any LLM/AI to get feedback: breach of authors' confidentiality rights and infringements of data protection.

Referee and editors must take responsibility.

Challenges and opportunities

Can (and does) help leveling the playing field for ESL (english as second language) authors

Could help finding appropriate journal/venue and tailor manuscript to that

Could help finding related literature (no hallucinations!) and help frame a given manuscript within the literature (especially for fields covered by arXiv)

Could help sifting through large volume of incremental work vs novel work.

Could help massively in finding the right editor and **especially the right expert referee** (reducing referee fatigue)- current tools are surprisingly limited-

Conclusions

- → AI can generate ideas, methods, and papers faster than any human.
→ But “more” science doesn’t mean *better* science.
- spam science
- → The *signal-to-noise ratio* collapses.

While we figure out what are the atoms of new scientific knowledge
and a new incentive system

It's a Red Queen race



symptoms, root causes, or consequences?

REFEREE FATIGUE

EDITOR FATIGUE

READERS FATIGUE

SALAMI SLICING

DELAYS

PAPER MILLS

PUBLISH OR PERISH

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SCALE!

NOW

Explosion of low quality papers

Misconducts harder to spot but amplified at scale

Literature becomes less reliable as a network of knowledge

Shallow understanding→ outsourced thinking

Acceleration and increased volume: information overflow but low signal to noise

Weakening the base trust of science

Easier to produce, harder to trust

AI can and should help
But the right incentives are
not there

THE INCENTIVES ARE THOSE THAT CREATED THE PROBLEM IN THE FIRST PLACE

2023 E-LIFE: PAY APC ON SUBMISSION. THE SUBMISSION GETS PUBLISHED ALONG WITH THE PEER REVIEW.

IN ECONOMICS THERE ARE SUBMISSION FEES.

